Extended Topic Abstract Form
(LAG-ASD-RKM-F-001)

EXTENDED TOPIC ABSTRACT FORM			
School	**School of Computer Studies**	Program Enrolled	**Computer Science**
Term/ Academic Year	**2nd Term/A.Y. 2025-2026**	Section	**BSCS231A**
Student's Name	**Arestado, Ted Carlo D.**		
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Proposed Title:	A Hybrid CSP-Heuristic Intelligent Scheduling System with Dynamic Priority Adjustment		

Key Components	**Content Writing**

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	Department	Research and Knowledge Management Office

Brief Background	• Current state of knowledge, consensus, present practice, or description of phenomena. There are no citations here; they are only in the introduction section/proper.	Research has shown that a significant percentage of university students vocalized that they feel stressed and anxious when their schedule is excessively loaded or poorly structured. The majority of them claim that disorderly academic demands divert their attention, lower their productivity, and make them feel suffocated continuously. Similar problems, as a result, can be traced to the professional field, where many employees have been identified as lacking the most fundamental time-management skills, such as task scheduling and activity prioritizing. These conditions reveal the common, profound problem of proper workload management, which is shared by both the educational and the working communities.
	• Identify the gap/s or describe the research problem/s. There are no citations here.	Present scheduling arrangements generally depend on inflexible, unchanging techniques that fail to accommodate any real-time changes in tasks or limitations. Because of this failure to modify, the schedules become so inefficient or inaccurate in a short time when there is a change of priorities. Quite a few of these systems also do not have features for dynamic decision-making, thus, they are less effective in complex surroundings. Therefore, users lose out on flexibility, responsiveness, and total productivity to a great extent. Such differences pinpoint the necessity for a more sensitive and clever solution to scheduling which would be able to manage the flow of changes without interruption.
Significance	• Articulate the significance of the proposed study.	This study is significant because it addresses the growing need for smarter, more adaptive personal scheduling systems in an era where poor time management is strongly linked to reduced productivity. Existing intelligent scheduling models primarily focus on workplace settings and often overlook the user's personal goals, changing priorities, and overall well-being. By introducing the Dynamic Goal Priority (DGP) algorithm, this research fills a critical gap in the literature by integrating dynamic priority adjustment with constraint-based scheduling. The system does not only assign tasks efficiently, it actively adapts to real-life changes, supports healthier time-management behaviors, and promotes well-being. As a result, the study offers a valuable contribution to fields such as artificial intelligence, personal productivity, and mental health–supportive technologies.

	• Explain in clear language your study's purpose.	The purpose of this study is to create an intelligent scheduling system that helps people manage their time better. Using the Dynamic Goal Priority (DGP) algorithm, the system automatically organizes a person's tasks based on what is most important, while still respecting fixed schedules such as school, work, or personal commitments. It adjusts in real time when goals change, new tasks appear, or urgent responsibilities arise. Simply put, the study aims to help users balance their priorities and improve productivity by giving them a flexible and personalized schedule that fits their daily lives.

Research Question/Objectives	• State the research questions (*Qualitative*) to be answered, OR research objectives (*often Quantitative*)	**Main Research Question** 1. How can a Dynamic Goal Priority (DGP) algorithm prioritize multiple user goals based on urgency and importance, and assign tasks into available free time without causing overload or conflict? **Supporting Research Questions** 1. How can urgency be computed from a goal's deadline and difficulty to determine its priority? 2. How can user-defined importance be used to break ties among goals with similar urgency? 3. How can the DGP algorithm generate and schedule generic tasks that support prioritized goals? 4. How can the algorithm avoid scheduling conflicts and prevent user burnout during task assignment? 5. How does the DGP approach compare to traditional scheduling approaches (e.g., deadline-based, priority-based)? **Main Objective** 1. To design and develop a Dynamic Goal Priority (DGP) algorithm that ranks user goals based on urgency and importance and schedules tasks into available free-time blocks in a human-centered manner. **Specific Objectives**

		1. To define a computation model for urgency based on goal deadline proximity and goal difficulty. 2. To incorporate user-defined importance as a tiebreaker for goals with equal urgency. 3. To generate generic tasks from each goal and assign them to available free-time blocks based on goal priority. 4. To integrate user-centered constraints (e.g., avoiding consecutive high-effort tasks) when assigning tasks. 5. To evaluate the DGP algorithm through simulations and compare its performance with standard scheduling methods.
Framework	• Articulate your theory/theoretical foundations that connect you to existing knowledge (*even hypotheses*) and your chosen research methods	**THEORETICAL FOUNDATIONS OF THE DGP ALGORITHM** The development of the Dynamic Goal Priority (DGP) algorithm is grounded in several existing theories and bodies of knowledge. These theories provide the conceptual foundation for how goals are prioritized, how urgency is interpreted, how tasks are scheduled, and why constructive research is the appropriate methodology for this study. **1. Priority Scheduling Theory** Priority Scheduling Theory explains how items (tasks, processes, or goals) can be arranged based on a priority value, with higher-priority items processed first. In computer science, priority scheduling is one of the fundamental CPU scheduling methods. **Connection to DGP:** ● The DGP algorithm also produces a **priority score**, but instead of processes, the DGP prioritizes **goals**. ● Like priority scheduling, DGP ranks items based on computed parameters—but expands the concept by combining **deadline urgency** and **user-defined importance**.

- DGP extends this theory by applying priority scheduling to **human goal management**, not machine processing.

Hypothesis Inspired by This Theory:

> If goals are assigned accurate priority scores, scheduling becomes more efficient and aligned with user needs.

2. Deadline-Based Scheduling (Earliest Deadline First Theory)

Earliest Deadline First (EDF) is a well-known scheduling algorithm where tasks with the nearest deadlines are executed first. It assumes deadlines represent urgency.

Connection to DGP:

- EDF is the foundation for **Layer 1 urgency computation**.
- DGP adopts the "closer deadline → higher urgency" principle.
- DGP modifies and improves EDF by adding **goal difficulty** to urgency computation, recognizing that some goals require more time and effort than others.

Hypothesis Inspired by This Theory:

> Urgency derived from deadlines and difficulty provides a more realistic measure of what should be prioritized first.

3. Weighted Scoring Models

Weighted scoring is used in decision science and project management to rank options by scoring factors and applying weights. It is commonly used in multi-criteria decision making (MCDM).

		Connection to DGP: • DGP uses weighted scoring to break ties between goals that have the same urgency. • User-defined **importance** becomes a weighted factor. • This ensures long-term important goals still receive time even when they are not deadline-pressured. **Hypothesis Inspired by This Theory:** If long-term important goals are considered through weighted importance, balanced progress becomes achievable. **4. Goal-Setting Theory (Locke & Latham)** Goal-Setting Theory states that: • Specific goals lead to higher performance. • People work better when goals are clear and structured. • Goals create commitment and direct attention to important tasks. **Connection to DGP:** • DGP focuses on **goals first**, not tasks. • The algorithm generates tasks based on the **goal's nature, difficulty, and time horizon**. • DGP aligns scheduling with long-term objectives, not daily fluctuations. **Hypothesis Inspired by This Theory:**

> A goal-based scheduling algorithm increases consistency and clarity in daily productivity.

5. Cognitive Load Theory & Human-Centered Scheduling

Cognitive Load Theory states that humans can only handle a limited amount of mental effort. Overloading reduces performance. Human-centered scheduling research extends this by emphasizing:

- Avoiding burnout
- Balancing workloads
- Including rest periods
- Preventing consecutive high-effort tasks

Connection to DGP:

- DGP assigns tasks **after** goal priority is determined, ensuring human-centered constraints:
    - No consecutive high-effort sessions
    - Respecting free-time limitations
    - Balancing multiple goals without burnout
- This forms the basis for realistic and sustainable scheduling in daily life.

Hypothesis Inspired by This Theory:

> A scheduling algorithm that respects human limitations will prevent overload and improve long-term adherence.

6. Constructive Research Method (Methodological Foundation)

		Constructive Research is used when the goal is to **design and develop an innovative solution**, usually a model, algorithm, or system. It includes: • Identifying a problem • Understanding the domain • Constructing the solution • Demonstrating the solution's effectiveness through testing **Connection to DGP:** • DGP is a **newly constructed scheduling algorithm**. • Its development follows the constructive process: 1. Identify the problem: people struggle to schedule multiple goals without conflict or burnout 2. Review existing knowledge: deadline scheduling, priority scoring, weighted scoring 3. Construct algorithm logic 4. Simulate performance using test scenarios 5. Analyze and compare results **Why This Method Fits:** • The study does not need human respondents at the algorithm construction stage. • It focuses on **designing, testing, evaluating, and validating a new algorithm**—exactly what constructive research is designed for.

		FINAL SYNTHESIS (Framework Statement) Bringing all these theories together: *The DGP algorithm is grounded in priority scheduling theory, enhanced by deadline-based urgency (EDF) and multi-criteria weighted scoring to determine goal priority. It embraces goal-setting theory by structuring user goals into actionable tasks and incorporates human-centered scheduling principles to avoid overload. The constructive research method provides a systematic approach to developing, simulating, and evaluating the algorithm's effectiveness in managing multiple goals within user free time.*
Methodology	• Discuss your methodology, specific data collection methods/techniques, sampling, respondents/ participants, recruitment process, instruments, and data analysis procedures.	**1. Research Design** This study uses the **Constructive Research Method**, which focuses on developing an innovative solution to a practical problem through the construction and evaluation of a new model or algorithm. The solution constructed in this study is the **Dynamic Goal Priority (DGP) algorithm**, which prioritizes user goals based on urgency (deadline × difficulty) and user-defined importance, then assigns generic tasks into free-time blocks while considering human constraints (e.g., workload balance, avoidance of consecutive high-effort tasks). Constructive research is appropriate because the study involves: 1. Identifying a real-life scheduling problem 2. Reviewing existing scheduling theories 3. Constructing the DGP algorithm 4. Simulating its performance

5. Evaluating the results against existing approaches

2. Data Collection Methods / Techniques

2.1. System-Generated Algorithm Data

Since this is an algorithm development study, no human data is required. Data is generated through **controlled simulation**, including:

- User goals
- Goal deadlines
- Goal difficulty levels
- User-defined importance values
- Generic tasks derived from goals
- User free-time schedule
- Constraints (fatigue limit, max high-effort sessions, etc.)

The algorithm outputs:

- Goal priority scores (Urgency → Importance tiebreaker)
- Task assignments per free-time block
- Conflict and fatigue adjustments
- Comparative scheduling performance

These outputs form the primary dataset for analysis.

2.2. Expert Validation

3–5 domain experts (IT professors or professionals) may be invited to review:

- Algorithm logic
- Priority formulas
- Scheduling constraints

They will use a rubric or checklist, not interviews or surveys.

3. Sampling

3.1. Sampling for Simulation Data

This study does **not involve human respondents**, so sampling applies only to the **test scenarios** fed into the algorithm.

Purposive sampling is used to design representative scenarios, including:

- Short-term urgent goals
- Long-term flexible goals
- High-difficulty goals
- Low-difficulty goals
- Limited free-time schedules
- Balanced schedules

Example scenario sets:

- **Scenario A:** 3 goals (urgent, medium, long-term)
- **Scenario B:** 5 goals with mixed difficulties

- **Scenario C:** Very limited or interrupted free time
- **Scenario D:** Heavy goals but large free time

This ensures the algorithm is evaluated under varied and realistic conditions.

3.2. Sampling for Validators

- **Purposive sampling** is used (IT experts familiar with scheduling or algorithm design).

4. Respondents / Participants

4.1. Primary Study

None.
The main algorithm development and evaluation do **not require human respondents** because:

- Inputs come from simulated goal/task sets
- Evaluation is computational and algorithmic
- No personal data is collected

4.2. Expert Validators

- 3–5 respondents
- Role: Validate algorithm logic
- No personal information collected

5. Recruitment Process

- Potential validators will be identified through faculty lists or professional networks.
- An invitation letter or email will be sent.
- Participation is voluntary.
- Validators will receive the algorithm description and rubric.
- No compensation, no personal data stored.

6. Research Instruments

6.1. DGP Algorithm Prototype (Main Instrument)

The core instrument is the **implemented algorithm**, which includes:

- Goal class (deadline, difficulty, importance)
- Priority computation module (Urgency → Importance)
- Task generator
- Scheduler (assigns tasks to free time)
- Constraint engine (fatigue, consecutive difficulty, conflicts)

6.2. Simulation Scenario Sheets

These contain:

- Defined goals
- Deadlines
- Difficulty levels
- Importance values

- Free-time blocks
- Expected behaviors

6.3. Validation Rubric

Used by validators to assess:

- Accuracy of priority formula
- Logical consistency
- Human-centered scheduling

7. Data Analysis Procedures

7.1. Algorithmic Analysis

After running simulation scenarios, outputs will be analyzed through:

- **Priority Accuracy**
  How well the algorithm ranks goals based on urgency × importance
- **Task Allocation Efficiency**
  How well tasks fit into free-time blocks
- **Conflict/Burnout Avoidance**
  Checks if constraints were successfully applied
- **Goal Progress Comparison**
  How much progress each goal achieves relative to its deadline

7.2. Comparative Analysis

DGP will be compared to simplified versions of existing scheduling methods:

- **Earliest Deadline First (EDF)**

		• **Priority Scheduling** • **Weighted Scoring Model** Metrics: • How balanced the resulting schedule is • Whether both short-term and long-term goals receive time • Avoidance of consecutive high-effort sessions • Utilization of free-time blocks **7.3. Descriptive Statistical Analysis** Results will be presented through: • Tables • Gantt-style visual schedules • Priority score breakdowns No inferential statistics are required because the data is algorithmic, not human-derived. **8. Ethical Considerations** • No personal or sensitive data is collected. • Optional validators provide only expert opinion, not personal data. • All simulations are artificial and contain no real user information.

		• Study adheres to academic integrity and transparency of algorithmic design.
Expected Outcomes / Deliverables	• Anticipated results or implications (Innovations, models, or tools that may be developed and the Application of findings)	**1. Development of the Dynamic Goal Priority (DGP) Algorithm** The primary delivery of this study is a fully developed **Dynamic Goal Priority (DGP) algorithm**. This algorithm is expected to: • Prioritize user goals based on **urgency (deadline × difficulty)** and **user-defined importance**. • Generate generic tasks based on each goal and assign them into the user's free-time schedule. • Apply human-centered constraints (avoiding overload, preventing consecutive high-effort tasks). • Optimize goal progress while maintaining realistic workload distribution. This represents an **innovative scheduling model** designed for multi-goal personal productivity. **2. Functional Prototype (Scheduling Tool)** A software prototype will be developed to demonstrate the DGP algorithm. The prototype is expected to: • Accept user-defined goals, deadlines, difficulty values, and importance levels.

- Display available free-time blocks.
- Automatically produce a realistic task schedule.
- Visualize the final schedule (e.g., timeline or Gantt-like output).
- Show the priority score of each goal and explain why certain goals were scheduled first.

This tool will serve as **proof of concept** for practical application.

3. Goal-Centered Scheduling Model

This study will contribute to a new **goal-oriented scheduling concept**, distinguishing itself from traditional task-based schedulers.
Expected contributions include:

- A clear model for computing **urgency** from deadlines and difficulty.
- A tie-breaking model using **importance scoring**.
- A human-centered scheduling logic that considers fatigue and workload balance.
- A structured flow for turning goals → generic tasks → scheduled blocks.

This advances personal productivity research by shifting the focus from "task lists" to "goal achievement."

4. Comparative Performance Insights

Through simulation, the research will provide performance comparisons between DGP and existing algorithms such as:

- Earliest Deadline First (EDF)

- Priority Scheduling
- Weighted Scoring Model

Expected findings:

- DGP is more balanced in distributing time across urgent and long-term goals.
- DGP prevents user overload better than deadline-only schedulers.
- DGP maintains goal progression more effectively than importance-only schedulers.
- DGP produces more human-realistic schedules than computing-based priority algorithms.

These findings will demonstrate the **practical superiority** of goal-centered scheduling.

5. Practical Implications

The study's results may be applied to:

- **Personal productivity apps**
- **Student planners / academic schedulers**
- **Task and goal management tools**
- **Wellness and fitness planning systems**
- **Any system balancing long-term and short-term goals**

The DGP model can be integrated as the core scheduling engine in future mobile or web-based applications.

6. Scholarly Contribution

The DGP algorithm and its supporting logic contribute to:

- Theoretical advancement in **goal-based scheduling**
- New frameworks for **urgency computation** using deadline × difficulty
- Constructive research output applicable to AI-based personal assistants
- An improved understanding of human-centered workload distribution

This supports future studies in algorithmic productivity, decision support systems, and automated planning.

7. Documentation and Technical Deliverables

The study will also produce:

- Algorithm documentation
- UML diagrams (Goal Class, Task Class, Scheduler class)
- Flowcharts for priority computation and task assignment
- Formula definitions for urgency, importance, and scheduling constraints
- User guide for the prototype tool

These materials ensure the algorithm can be replicated, evaluated, and extended by future researchers.

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	Department	Research and Knowledge Management Office

Research Agenda/Sustainable Development	• Addressed Research Agenda/Sustainable Development Goals:	SDG 3 – Good Health and Well-being ● Takes well-being into account when scheduling, preventing stress and burning out. ● Encourages healthy habits like exercise and adequate rest. SDG 4 – Quality Education ● Optimizes time for study and learning-related tasks. ● Helps students manage coursework efficiently and support lifelong learning. SDG 8 – Decent Work and Economic Growth ● Enhances productivity and time management for work-related tasks. ● Allows better balance of work and personal life, improving efficiency and reducing stress. SDG 9 – Industry, Innovation, and Infrastructure ● Introduces an innovative algorithmic solution for personal productivity and task management. ● It can inspire smart scheduling tools and AI-driven productivity platforms. SDG 10 – Reduced Inequalities ● Helps users with heavy workloads manage time efficiently. ● Reduces time-management inequality, giving users equal opportunity to balance work, education, and well-being. SDG 12 – Responsible Consumption and Production ● Encourages efficient use of personal time, minimizing wasted effort. ● Promotes intentional, goal-oriented use of energy and attention.

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Keywords	• Include three to five keywords related to the proposed study.	1. Dynamic Goal Priority (DGP) 2. Goal-Based Scheduling 3. Urgency Computation 4. Importance Scoring 5. Heuristic Scheduling 6. Task Allocation 7. Human-Centered Scheduling 8. Multi-Goal Management
References	• List of key references used in APA 7th Edition format	Lovin, D., & Bernardeau-Moreau, D. (2022). Stress among Students and Difficulty with Time Management: A Study at the University of Galați in Romania. Social Sciences, 11(12), 538. https://doi.org/10.3390/socsci11120538 Terefe, S., et al. (2023). Time management practice and associated factors among employees working in public health centers, Dabat District, Northwest Ethiopia. BMC Health Services Research, 23(1).

TOPIC EVALUATION COMMITTEE The topic abstract has been thoroughly reviewed by the Topic Evaluation Committee.				
Designation	**Names**	**Signature**	**Date**	**Remarks**
Panel Chair (Subject Matter Expert)				
Expert Member 1 (Statistician)				
Expert Member 2 (Methodologist)				

Research Adviser				
Research Instructor				